

Generation Interconnection System Impact Study Report

For Queue Project Q16-0600

Lansing 100 kV

July 29, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0600. The Interconnection Customer proposes a 150.0 MW solar photovoltaic facility interconnecting at the Lansing 100 kV substation in the Illinois area. The request was received on March 15, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 276 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,506,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DYCKESVILLE-LANSING 100.0 kV	SAND RIDGE-CASTLEAFB 100.0 kV	152.4	133	41.0
SAND RIDGE-CASTLE AFB 100.0 kV line	DYCKESVILLE SUBSTATION-LANSING 100.0 kV	125.7	134	26.6
SHROUD-TOWER AUTOMOTIVE 100.0 kV	SHROUD-TOWER AUTOMOTIVE 100.0 kV Ckt 1	117.0	153	8.5
TOWER AUTOMOTIVE-SHROUD 100.0 kV Ckt 1	TOWER AUTOMOTIVE-SHROUD 100.0 kV Line 2	116.1	154	8.5
CASTLE AFB-DYCKESVILLE 100.0 kV	CASTLE AFB-SAND RIDGE 100.0 kV	111.6	135	12.4
DYCKESVILLE-EAST JACKSON 1&2 SUBSTATION 100.0 kV	SHOREHAM POWER STA-CASTLE AFB SUBSTATION 100.0 kV	107.1	109	35.9
SHOREHAM POWER STA-CASTLE AFB 100.0 kV	BLACKFOOT-WEST DANVILLE 15 230.0 kV	105.6	137	14.2
FRANZ HOLMES-SAND RIDGE 100.0 kV L	DYCKESVILLE-LANSING 100.0 kV	105.4	138	12.0
WILDWOOD-SAND BAR 100.0 kV #2	WILDWOOD-SAND BAR 100.0 kV Cir. 1	105.2	144	6.0
WILDWOOD-SAND BAR 100.0 kV #1	WILDWOOD-SAND BAR 100.0 kV Line 2	104.7	145	6.0
SAND BAR-ROBBINS COMMUNITY POWER LLC 100.0 kV Ckt 1	ROBBINS COMMUNITY POWER LLC-SAND BAR 100.0 kV Line 2	102.8	145	5.8
TERRYVILLE-SYCAMORE COGEN 100.0 kV	SWEATT SUBSTATION-LAWRENCE HILL 100.0 kV	100.6	137	14.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DYCKESVILLE-LANSING 100.0 kV	Rebuild line with higher-capacity conductor	\$1,254,000	60% (\$757,000)
SAND RIDGE-CASTLE AFB 100.0 kV	Rebuild line with higher-capacity conductor	\$1,346,000	90% (\$1,212,000)
SHROUD-TOWER AUTOMOTIVE 100.0 kV	Rebuild line with higher-capacity conductor	\$6,414,000	68% (\$4,382,000)
TOWER AUTOMOTIVE-SHROUD 100.0 kV Ckt 1	Rebuild line with higher-capacity conductor	\$3,988,000	68% (\$2,724,000)
CASTLE AFB-DYKESVILLE 100.0 kV	Rebuild line with higher-capacity conductor	\$2,061,000	100% (\$2,061,000)
DYCKESVILLE-EAST JACKSON 1&2 100.0 kV	Reconductor line	\$707,000	62% (\$440,000)
SHOREHAM POWER STA-CASTLE AFB 100.0 kV	Reconductor line	\$837,000	78% (\$652,000)
FRANZ HOLMES-SAND RIDE 100.0 kV	Reconductor line	\$288,000	100% (\$288,000)
SAND BAR SUBSTATION-WILDWOOD 100.0 kV #2	Reconductor line	\$490,000	62% (\$303,000)
WILDWOOD-SAND BAR 100.0 kV	Reconductor line	\$481,000	62% (\$297,000)
ROBBINS COMMUNITY POWER LLC-SAND BAR 100.0 kV Ckt 1	Reconductor line	\$853,000	62% (\$527,000)
SYCAMORE COGEN-TERRYVILLE 100.0 kV Line	Reconductor line	\$863,000	100% (\$863,000)
	Total allocated to project		\$14,506,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.